

Education

- 2015-2020 **Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering**
Indian Institute of Technology Madras, Chennai, India CGPA: 8.78
- 2015 **XII - Karnataka Board, KLE Society's Independent PU College, Bangalore** 97.30 %
- 2013 **X - ICSE, B P Indian Public School, Bangalore** 96.33%

Professional Experience

- Software Engineer, Level IV, Google LLC, New York**
- Dec 2022 - Present **Software Engineer, Level IV, Google India Pvt Ltd, Bangalore**
- Aug 2020 - Nov 2022 { Working on Keep, a notetaking editor in Google Workspace.
 { Developing intelligent features for the Google Workspace Editors (Docs, Slides, Keep, etc) using my expertise on the products' client-side software, supporting tools and libraries, and natural language processing infrastructure.
 { Using cutting-edge frontend tools like Web Assembly and Emscripten, and Google-internal technologies like j2Cl, client-side cross-platform frameworks and build systems, to develop user-facing features such as spellcheck in encrypted documents for five languages and writing style suggestions for English text.
 { Formulating technical designs for independent end-to-end problems, driving cross-team collaboration, upholding software reliability practices, technical-debt resolution and documentation, and proactively identifying areas of future work.
 { Guiding junior engineers on programming and software design tasks to enable timely delivery of products to customers.
- May 2019 - July 2019 **Software Engineering Intern, Google India Pvt Ltd, Bangalore**
- Worked on the Editors client-side software infrastructure to develop a user interface with control options to undo or provide feedback on the correction and a logging framework, for the Google Docs text auto-correction feature.
- May 2018 - July 2018 **Research Intern, Big Data Experience Labs, Adobe Research, Bangalore**
- Developed a mobile application for Text to Scene Conversion in Augmented Reality, based on novel research techniques for prediction of three-dimensional object sizes and positions from textual features.

Research Experience

- Sep 2019 - May 2020 **Paraphrase Generation with a Bilingual Model and Continuous Embeddings** *Master's Thesis, Language Technologies Institute, Carnegie Mellon University*
- Machinated a novel technique for paraphrase generation using the [von Mises-Fisher \(vMF\) Loss](#) on a transformer network, and showed that it produces superior paraphrases as compared to the log-likelihood model by employing bilingual data to induce zero-shot paraphrasing, guided by [Prof. Yulia Tsvetkov](#).
- May 2017 - July 2017 **Cognitive Approach to Natural Language Processing** *Research Intern, Department of Computer Science and Automation, Indian Institute of Science (IISc), Bangalore*
- Developed a cognitive text parser that combines syntactic and semantic approaches, to process textual data into cognitive structural representations, to be used as a feature extractor for downstream NLP tasks, and demonstrated the correlation of the extracted cognitive features with semantic and syntactic text features, guided by [Prof. Veni Madhavan](#).

Publications and Patents

- [Publication and Poster] **Improving the Diversity of Unsupervised Paraphrasing with Embedding Outputs (Paper, Poster)**
Monisha Jegadeesan, Sachin Kumar, John Wieting, Yulia Tsvetkov
 In [Workshop on Multilingual Representation Learning](#),
 The 2021 Conference on Empirical Methods in Natural Language Processing ([EMNLP 2021](#))
- [Publication and Poster] **Adversarial Demotion of Gender Bias in Natural Language Generation (Paper, Poster)**
Monisha Jegadeesan
 In [ACM CODS-COMAD 2020](#) - Young Researchers' Symposium
- [Poster] **ARComposer: Authoring Augmented Reality Experiences through Text (Poster)**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 In ACM User Interface Software and Technology Symposium 2019 ([ACM UIST 2019](#))
- [Filed Patent] **Visualizing Natural Language through 3D Scenes in Augmented Reality**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 Filed at the US PTO (Application Number: 16/247,235)

Projects

Real-Time Customer Support Chatbot with Scalable AI Pipelines (2024)

Designed and deployed a high-performance AI-powered chatbot leveraging scalable AI pipelines to reduce human agent workload by 35% and provide instant query responses. Architected and maintained retrieval-augmented generation (RAG) pipeline utilizing vector databases and LLM integration for real-time interaction. Managed cloud infrastructure to optimize GPU resources ensuring reliability and scalability for customer support workloads.

Tech: LangChain, OpenAI API, ChromaDB, Python, GPU clusters, Cloud Infrastructure

Predictive Maintenance Pipeline for Industrial AI Workloads (2024)

Engineered and maintained scalable machine learning pipelines for predictive maintenance using IoT sensor data to forecast machine failures. Designed infrastructure to support high-throughput feature extraction and model training on GPU-enabled clusters. Achieved a 28% reduction in unexpected downtime by deploying reliable Random Forest models leveraging automated, high-performance computing resources for continuous inference.

Tech: Python, Random Forest, IoT Sensor Data, GPU clusters, High-Performance Computing

Call Center Performance Analytics with Scalable Cloud Infrastructure (2024)

Developed and deployed a cloud-based analytics dashboard to monitor and improve call center KPIs supporting data-driven decision making. Designed scalable ETL pipelines with SQL and optimized data models using DAX to process large volumes of call logs efficiently. Enabled real-time visualization on Power BI with high reliability, driving a 22% reduction in wait times and 13% SLA compliance improvement through operational insights.

Tech: SQL, Power BI, DAX, Excel, Cloud Infrastructure, ETL Pipelines

Customer Sentiment Analysis Pipeline for E-commerce NLP Workloads (2024)

Implemented scalable NLP pipelines for sentiment classification on large-scale e-commerce review datasets, hosted on cloud infrastructure optimized for high-performance computing. Automated preprocessing, feature extraction, and model training workflows ensuring reliability in production using GPU acceleration. Delivered 89% classification accuracy, enabling actionable insights for product improvements via robust AI pipelines designed for operational efficiency.

Tech: Python, NLTK, TF-IDF, Logistic Regression, Cloud Infrastructure, GPU clusters

About :

I specialize in designing, deploying, and maintaining scalable, high-performance ML models and AI pipelines to solve real-world problems. Proficient in Python, scikit-learn, TensorFlow/PyTorch, and experienced with end-to-end ML workflows including data preprocessing, feature engineering, model optimization, and performance evaluation. Skilled in designing and managing GPU/TPU clusters, high-performance computing systems, and cloud infrastructure to support AI workloads. Experienced in deploying models into production environments, integrating them into applications via APIs, and collaborating closely with ML engineers, data scientists, and DevOps teams to ensure scalable, reliable, and efficient AI solutions that drive measurable business impact.

